



CS 540 Project Guidelines and Rubric

Competencies

In this project, you will demonstrate your mastery of the following competencies:

- Gather data required for informing organizational decision-making processes
- Create data visualizations appropriate for revealing key data insights
- Use data visualizations in the support of a data-driven narrative

Overview

In today's world, organizations rely on data literacy and visualization skills to make strategic decisions. These decisions impact profits, efficiency, and innovation. For the first part of this project, you will perform exploratory data analysis techniques on provided or chosen data sets to find patterns and trends. For the second part, you will perform explanatory data analysis techniques to present your findings to two separate groups of stakeholders. Your role is to turn raw data into helpful information that supports organizational decision making. This project prepares you for careers in business intelligence, consulting, and data analysis. Learning these skills will improve decision making and help shape the direction of any organization.

Scenario

As a newly hired data analyst, your job is to help your company understand the trends in the available data. To do this, you will need to rely on reputable, publicly available data sources. You'll be expected to gather and organize this data into a clear and usable format that can reveal patterns in behavior and preferences. From there, you'll explore the data to uncover meaningful insights and use visualizations to make those insights easy to understand. Your work will support key decision making, so you'll present your findings to two groups of stakeholders at the company in a way that's accessible and relevant to each audience.

Directions

For this project, choose at least two data sets from a reputable data source in your field of interest or from one of the data providers listed in Codio and customize the resulting data-driven narrative for two groups of stakeholders. You will complete your work using various tools available within the Codio environment. Your submission may include written explanations and screenshots as needed to communicate your process and findings clearly. When you are ready to begin your work, access Codio by clicking the title of the next task.

When the assignment is complete, download the HTML version of your project by clicking on **File**, then **Download As**, then **HTML (.html)** from the Jupyter menu bar. Save the file to your local machine.

***Note:** if the code block was not run in the Notebook, it may not render in the HTML output of the Notebook. To fix missing blocks, run the code block, then download as HTML.*

Part One: Data Preparation and Visualizations

1. Perform **exploratory data analysis** to identify key data points.

2. Identify **relevant data** from your chosen data sets. Include:
 - A. A citation of your data sources
3. Evaluate the **quality** of your chosen data.
4. Apply appropriate **data preprocessing techniques**.
5. **Blend multiple data sets** into a single integrated collection.
6. Choose **appropriate visualization types** to reveal key trends within the data collection.
7. Integrate **multiple data attributes** into visualizations using appropriate techniques.
8. Derive **key insights** from your visualizations.
9. Use **multi-level data modeling** practices to refine your visualizations. Include:
 - A. A high-level visualization that provides a broad overview of the data
 - B. A detailed, insight-focused visualization that highlights the key findings

Part Two: Data-Driven Narrative

10. Describe the **purpose and audiences** for your data-driven narrative. Include:
 - A. Descriptions of the two groups of stakeholders
11. Develop a compelling **data-driven narrative** using storytelling principles.
12. Evaluate the **effectiveness of your chosen visualizations** in communicating your data insights.
13. Customize your **integrated visualizations** to help each audience better understand the insights. Include:
 - A. A justification of how your customization choices improve visualization clarity and highlight key insights.

What to Submit

Data Preparation and Visualizations

Download your Jupyter Notebook as an HTML file. Review the file to ensure the code and data visualizations are displayed. Submit the HTML file.

Data-Driven Narrative

Submit your completed data-driven narrative as a 4- to 6-page Microsoft Word document with double-spacing, 12-point Times New Roman font, and one-inch margins.

AI Usage

If you use gen AI tools to support your work on this assignment, be sure to follow [these AI usage guidelines](#). You must acknowledge your use of these tools in your work. Guidelines on how to cite AI tools can be found in [this Shapiro Library guide](#).

Project Rubric

Criteria	Exceeds Expectations (100%)	Meets Expectations (90%)	Partially Meets Expectations (70%)	Does Not Meet Expectations (0%)	Value
Part One: Exploratory Data Analysis	Exceeds expectations in an exceptionally clear, organized, insightful, or sophisticated manner	Performs exploratory data analysis to identify key data points	Shows progress toward meeting expectations, but with errors or omissions	Does not attempt criterion	6
Part One: Relevant Data	Exceeds expectations in an exceptionally clear, organized, insightful, or sophisticated manner	Identifies relevant data from the chosen data sets, including a citation of the data sources	Shows progress toward meeting expectations, but with errors or omissions	Does not attempt criterion	6
Part One: Quality	Exceeds expectations in an exceptionally clear, organized, insightful, or sophisticated manner	Evaluates the quality of the chosen data	Shows progress toward meeting expectations, but with errors or omissions	Does not attempt criterion	6
Part One: Data Preprocessing Techniques	Exceeds expectations in an exceptionally clear, organized, insightful, or sophisticated manner	Applies appropriate data preprocessing techniques	Shows progress toward meeting expectations, but with errors or omissions	Does not attempt criterion	6
Part One: Blend Multiple Data Sets	Exceeds expectations in an exceptionally clear, organized, insightful, or sophisticated manner	Blends multiple data sets into a single integrated collection	Shows progress toward meeting expectations, but with errors or omissions	Does not attempt criterion	6
Part One: Appropriate Visualization Types	Exceeds expectations in an exceptionally clear, organized, insightful, or sophisticated manner	Chooses appropriate visualization types to reveal key trends within the data collection	Shows progress toward meeting expectations, but with errors or omissions	Does not attempt criterion	7.5
Part One: Multiple Data Attributes	Exceeds expectations in an exceptionally clear, organized, insightful, or sophisticated manner	Integrates multiple data attributes into visualizations using appropriate techniques	Shows progress toward meeting expectations, but with errors or omissions	Does not attempt criterion	7.5

Criteria	Exceeds Expectations (100%)	Meets Expectations (90%)	Partially Meets Expectations (70%)	Does Not Meet Expectations (0%)	Value
Part One: Key Insights	Exceeds expectations in an exceptionally clear, organized, insightful, or sophisticated manner	Derives key insights from the visualizations	Shows progress toward meeting expectations, but with errors or omissions	Does not attempt criterion	7.5
Part One: Multi-level Data Modeling	Exceeds expectations in an exceptionally clear, organized, insightful, or sophisticated manner	Uses multi-level data modeling practices to refine the visualizations, including a high-level visualization that provides a broad overview of the data and a detailed, insight-focused visualization that highlights the key findings	Shows progress toward meeting expectations, but with errors or omissions	Does not attempt criterion	7.5
Part Two: Purpose and Audiences	Exceeds expectations in an exceptionally clear, organized, insightful, or sophisticated manner	Describes the purpose and audiences for the data-driven narrative, including descriptions of the two groups of stakeholders	Shows progress toward meeting expectations, but with errors or omissions	Does not attempt criterion	7.5
Part Two: Data-Driven Narrative	Exceeds expectations in an exceptionally clear, organized, insightful, or sophisticated manner	Develops a compelling data-driven narrative using storytelling principles	Shows progress toward meeting expectations, but with errors or omissions	Does not attempt criterion	7.5
Part Two: Effectiveness of Visualizations	Exceeds expectations in an exceptionally clear, organized, insightful, or sophisticated manner	Evaluates the effectiveness of the chosen visualizations in communicating the data insights	Shows progress toward meeting expectations, but with errors or omissions	Does not attempt criterion	7.5
Part Two: Integrated Visualizations	Exceeds expectations in an exceptionally clear, organized, insightful, or sophisticated manner	Customizes the integrated visualizations to help the audiences better understand the insights, including a justification of how the customization choices improve visualization clarity and highlight key insights	Shows progress toward meeting expectations, but with errors or omissions	Does not attempt criterion	7.5

Criteria	Exceeds Expectations (100%)	Meets Expectations (90%)	Partially Meets Expectations (70%)	Does Not Meet Expectations (0%)	Value
Clear Communication	Exceeds expectations with an intentional use of language that promotes a thorough understanding	Consistently and effectively communicates in an organized way to a specific audience	Shows progress toward meeting expectations, but communication is inconsistent or ineffective in a way that negatively impacts understanding	Shows no evidence of consistent, effective, or organized communication	10
Total:					100%